

IB AHL & AIHL

HOW TO INTEGRATE

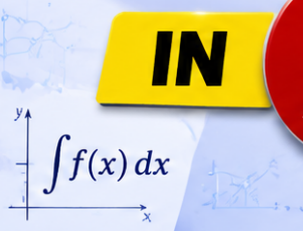
USING SUBSTITUTION

IN

5

STEPS

IB AHL 5.16


$$\int f(x) dx$$



$$\int f(x) dx$$



- 1 CHOOSE u** Let $u = g(x)$
- 2 DIFFERENTIATE** $du = g'(x) dx$
- 3 SUBSTITUTE** Write the integral in terms of u
- 4 INTEGRATE** Integrate with respect to u
- 5 BACK SUBSTITUTE** Replace u with $g(x)$ and simplify

Example:

$$\int (2x + 3)^4 dx \Rightarrow \frac{1}{10} (2x + 3)^5 + C$$



*Clear
Concepts
Better
Marks!*



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AHL 5.16

Content	Guidance, clarification and syllabus links
1. Integration by substitution.	On examination papers, substitutions will be provided if the integral is not of the form $\int kg'(x)f(g(x))dx$. Link to: integration by substitution (SL5.10).
2. Integration by parts.	Examples: $\int x \sin x dx$, $\int \ln x dx$, $\int \arcsin x dx$
3. Repeated integration by parts.	Examples: $\int x^2 e^x dx$ and $\int e^x \sin x dx$.

1. Integration by substitution:

In this type of Integrals there will be two terms. You have to substitute one of them as "u" whose derivative is the other term. You may require some adjustment in the other term.



$$(1) \int \frac{6x}{3x^2 + 7} dx$$

$$\text{let } u = 3x^2 + 7$$

$$\frac{du}{dx} = 6x$$

$$(2) \int (2x + 3) (\underline{x^2 + 3x})^{3/2} dx$$

$$\text{let } u = x^2 + 3x$$

$$\frac{du}{dx} = 2x + 3$$

$$(3) \int \frac{3x^2 + 4x}{\sqrt{\underline{\underline{x^3 + 2x^2}}}} dx$$

$$\text{let } u = x^3 + 2x^2$$

$$\frac{du}{dx} = 3x^2 + 4x$$



$$(1) \int \frac{6x}{3x^2 + 7} dx$$

$$\text{Let } u = 3x^2 + 7$$

$$\therefore \frac{du}{dx} = 6x$$

$$\therefore \int \frac{\left(\frac{du}{\cancel{dx}}\right)}{u} \cancel{dx}$$

$$= \int \frac{du}{u}$$

$$= \ln(u) + c$$

$$= \ln(3x^2 + 7) + c$$

$$(2) \int (2x + 3)(x^2 + 3x)^{3/2} dx$$

$$\text{Let } u = x^2 + 3x$$

$$\therefore \frac{du}{dx} = 2x + 3$$

$$\therefore \int \left(\frac{du}{\cancel{dx}}\right) (u)^{3/2} \cancel{dx}$$

$$= \int u^{3/2} du$$

$$= \frac{u^{5/2}}{5/2} + c$$

$$= \frac{2}{5} (x^2 + 3x)^{5/2} + C$$

$$(3) \int \frac{3x^2 + 4x}{\sqrt{x^3 + 2x^2}} dx$$

$$\text{Let } u = x^3 + 2x^2$$

$$\therefore \frac{du}{dx} = 3x^2 + 4x$$

$$\therefore \int \frac{\left(\frac{du}{\cancel{dx}}\right)}{\sqrt{u}} \cancel{dx}$$

$$= \int u^{-1/2} du$$

$$= \frac{u^{1/2}}{1/2} + C$$

$$= 2\sqrt{u} + C$$

$$= 2\sqrt{x^3 + 2x^2} + C$$



Past paper Question:

[Maximum mark: 7]

(a) Find $\int 4xe^{x^2-9} dx$. [4]

(b) Find $f(x)$, given that $f'(x) = 4xe^{x^2-9}$ and $f(3) = 7$. [3]

(a) $\int 4xe^{x^2-9} dx$.

$$\text{Let } u = x^2 - 9 \therefore \frac{du}{dx} = 2x$$

$$\int 4xe^{x^2-9} dx = 2 \int 2xe^{x^2-9} dx = 2 \int \frac{du}{dx} e^u dx = 2 \int e^u du = 2e^u + c$$

$$f(x) = 2e^{x^2-9} + c$$



(b) Find $f(x)$, given that $f'(x) = 4xe^{x^2-9}$ and $f(3) = 7$.

Integrating w.r.t. 'x', we get

$$f(x) = \int 4xe^{x^2-9} dx = 2e^{x^2-9} + c$$

$$f(3) = 7$$

$$\therefore 7 = 2e^{3^2-9} + c \Rightarrow 7 = 2e^0 + c \Rightarrow 7 = 2 + c$$

$$\therefore c = 5$$

$$\therefore f(x) = 2e^{x^2-9} + 5$$



Practice Quiz: Paper-1:

1. Evaluate the definite integral $\int_0^2 x\sqrt{4-x^2} dx$ using the substitution $u = 4 - x^2$.

A. $\frac{4}{3}$

B. $\frac{8}{3}$

C. 4

D. $\frac{16}{3}$

2. Find the indefinite integral $\int x^3\sqrt{x^2+1} dx$ using the substitution $u = x^2 + 1$.

A. $\frac{1}{5}(x^2+1)^{5/2} + \frac{1}{3}(x^2+1)^{3/2} + C$

B. $\frac{2}{5}(x^2+1)^{5/2} - \frac{2}{3}(x^2+1)^{3/2} + C$

C. $\frac{1}{3}(x^2+1)^{3/2} - \frac{1}{5}(x^2+1)^{5/2} + C$

D. $\frac{1}{5}(x^2+1)^{5/2} - \frac{1}{3}(x^2+1)^{3/2} + C$



3. Evaluate the indefinite integral $\int \frac{x}{\sqrt{1-x^4}} dx$.

A. $\frac{1}{2} \arcsin(x^2) + C$

B. $\arcsin(x^2) + C$

C. $\frac{1}{2} \ln|1-x^4| + C$

D. $-\frac{1}{2} \sqrt{1-x^4} + C$

4. Evaluate the definite integral $\int_0^1 \frac{x}{\sqrt{x+1}} dx$.

A. $\frac{4+2\sqrt{2}}{3}$

B. $\frac{2-\sqrt{2}}{3}$

C. $\frac{4-2\sqrt{2}}{3}$

D. $\frac{2\sqrt{2}-2}{3}$

5. Find the indefinite integral $\int \frac{e^x}{e^{2x}+4} dx$.

A. $\arctan\left(\frac{e^x}{2}\right) + C$

B. $\frac{1}{2} \arctan\left(\frac{e^x}{2}\right) + C$

C. $\frac{1}{4} \arctan\left(\frac{e^x}{2}\right) + C$

D. $\frac{1}{2} \ln(e^{2x} + 4) + C$

Note: Please solve these 5 quiz questions and let me know your answers or if you need help, on my WhatsApp given on next page.



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End of Lesson

Contact me on:  **9326363793**

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